

N L Subramanya Tejas

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Experience

American Express (via Cognizant)

Dec 2020 – Present

Engineer II

Bengaluru, India

Enterprise Search Platform

- Led a 6-month architectural migration that extracted Global Search from a shared enterprise MFE application into a dedicated OneApp MFE, establishing independent release cycles and ownership of configuration and enterprise integrations.
- Designed and provisioned the new root module and supporting infrastructure, resolving runtime dependencies and migrating analytics, experimentation, feedback, and chatbot integrations to the dedicated tenancy.
- Designed an Akamai rewrite strategy to seamlessly serve the new Search application behind the existing user-facing URL, preserving the established user experience without a browser-level redirect.
- Eliminated a 10-second analytics tracking blind spot by reconfiguring script execution order across Adobe Analytics and experimentation integrations.
- Led frontend delivery of semantic typeahead across enterprise Search Box deployments, evolving the experience from static keyword suggestions to intent-aware suggestions powered by hybrid lexical, vector, and LLM-based retrieval.
- Implemented 200ms debouncing and in-flight request cancellation for typeahead API requests, reducing unnecessary network traffic and preventing stale responses during rapid user input.
- Enabled context-aware search by identifying the consuming MFE/tenancy and passing this context to backend services, allowing suggestions to be scoped to the user's current application context.
- Designed deterministic GUID-based test and control bucketing for phased LLM feature rollouts, coordinating model governance approvals, tenancy-level enablement, and deployments across independently managed consumers.
- Built first-party suggestion analytics capturing search terms, displayed suggestions, clicked suggestion index, and consuming tenancy, enabling KPI measurement and downstream reporting through the enterprise data pipeline.
- Partnered with Product, Analytics, Marketing Operations, and engineering teams to define typeahead KPIs, telemetry requirements, rollout strategy, and adoption across consuming applications.
- Broke down PI-planned frontend requirements into user stories, estimated and delegated implementation work, and mentored engineers joining the team throughout architecture, development, and delivery.
- Uplifted the Search Box to the latest American Express Design Language System (DLS), improving accessibility and aligning the experience with current enterprise design standards.
- Architected and established the frontend foundation for a new internal Search as a Service workflow platform, provisioning cloud services, configuring SSO, and establishing Dev and QA environments.
- Designed and implemented a scalable OneApp MFE architecture with a root module, shared container for common layouts, and independently composed child modules, with RBAC designed for granular access to individual workflows.
- Defined the initial Search Data Review workflow to streamline manual review of Typeahead keywords, including accepting, rejecting, and replacing keywords and reviewing associated centroid data; created AI-assisted HTML prototypes to accelerate designer handoff and align workflows with DLS standards.
- Modernized frontend engineering practices across UI modules by establishing a release-oriented OneFlow branching strategy with `main` kept continuously releasable, while refactoring legacy code, modernizing class-based components, removing code smells, and improving performance.
- Authored migration documentation and architectural guidelines to support future OneApp tenancy transitions.

Partner Servicing Portal – Enterprise Onboarding Platform

- Developed an enterprise onboarding application used by partner banks to onboard and launch card products within the American Express ecosystem.
- Implemented complex multi-step workflows, product lifecycle dashboards, and card art management capabilities supporting thousands of assets.
- Reduced page load time by 30% through code-splitting and performance profiling, and improved accessibility scores by 45% to meet WCAG standards.
- Reduced reported security vulnerabilities by 60% through secure coding practices and PRISMA scanning; migrated the test stack from Enzyme to React Testing Library.
- Assumed feature ownership during tech lead absence, coordinating with business and engineering stakeholders to drive requirements and ensure timely delivery.

Projects

ForexFix – Travel FX Intelligence Platform | Next.js 15, TypeScript, Supabase, Gemini LLM, Tailwind CSS

- Designed and built a full-stack travel FX intelligence application that combines live interbank exchange rates, rule-based fee and spread modeling, and LLM-generated strategy recommendations.
- Architected a deterministic financial calculation layer separately from LLM-generated insights, ensuring numerical accuracy while using the model for strategy generation and contextual recommendations.
- Integrated live exchange-rate data with configurable fee and spread rules, enabling users to compare card, ATM, cash, and other FX strategies based on their travel context.
- Implemented Supabase-backed caching and TTL-based data freshness controls to reduce external API and LLM usage while maintaining timely rate and strategy information.
- Built the application using Next.js App Router, React Server Components, API routes, and TypeScript; deployed the production application on Vercel with environment-specific configuration.
- Leveraged AI-assisted development workflows to accelerate implementation and iteration while maintaining architectural control over application design, data flow, and model boundaries.

Technical Skills

Frontend Engineering: JavaScript (ES6+), TypeScript, React, Next.js, Micro-Frontend Architecture (OneApp), Redux, Context API, Performance Optimization, Accessibility (WCAG), Design Systems
Backend & Cloud: Node.js, REST APIs, API Integration, Authentication (SSO, JWT), Supabase, Cloud Services
Testing & Quality: Jest, React Testing Library, Cucumber (BDD), Unit & Integration Testing, SonarQube, PRISMA
AI & Search: LLM Application Development, Prompt Engineering, Context Engineering, RAG, Semantic Search, Hybrid Search, Embeddings, LangChain, LangSmith, Flowise
DevOps & Observability: CI/CD (GitHub Actions, Jenkins), Docker, Splunk, OpenSearch

Education

Atria Institute of Technology (VTU)	2016 – 2020
<i>B.E. in Computer Science & Engineering</i>	<i>Bengaluru, India</i>